

Financial News Sentiment Analysis: TF-IDF, FinBERT and Qwen2.5-1.5B Compared, and What Sentiment Does and Does Not Measure

A comparison of lexical, domain-pretrained and general-purpose language models, with a data examination and a critical evaluation of sentiment as a market signal

Abstract

Sentiment analysis of financial news is widely used to monitor markets, but it is rarely examined critically. This report does both. It first describes the data and method: 4,846 English financial sentences labelled positive, neutral or negative (59% neutral, 28% positive, 12% negative), divided into a stratified 80/20 training and test split. Because of the imbalance, a trivial always-neutral classifier reaches about 59% accuracy but only 0.25 Macro F1. Three classifiers are then compared on the 970 test sentences: a TF-IDF + logistic regression baseline, FinBERT used as released, and Qwen2.5-1.5B-Instruct prompted zero-shot. FinBERT is clearly strongest (accuracy 87.22%, Macro F1 0.86), while TF-IDF and Qwen tie at 74.95% accuracy with very different error profiles: TF-IDF leans towards the majority class, Qwen is cautious about negatives, and FinBERT over-interprets neutral events (42 of the 43 sentences on which only FinBERT failed are neutral). Finally, the report asks whether a high score on such a benchmark means that sentiment analysis reflects the market: whether the labels capture investor attitude rather than text tone (fidelity), whether a classified headline arrives before prices have already moved (speed), and whether all headlines deserve equal weight (weighting). Drawing on prior research, it argues that each of these gaps is real, and it proposes an upgraded design that adds current frontier LLMs, fair fine-tuned baselines, hybrid routing and market-linked evaluation.

Keywords: financial sentiment analysis; FinBERT; large language models; text classification; class imbalance; market efficiency

1. Introduction

Financial markets move on information, and much of that information arrives as text: news headlines, earnings releases and analyst notes. Automated sentiment analysis turns such text into a numerical signal that can be used to monitor risk, screen investments or build trading features. Financial language is, however, a difficult domain. Words such as “margin”, “exposure” or “cut” carry polarity only in context, general-purpose sentiment dictionaries misclassify many financial terms (Loughran & McDonald, 2011), and many headlines announce events (an acquisition, a plant opening, a negotiation) that are factual rather than emotional.

This report examines financial sentiment analysis at two levels. The first is technical: how well do three modelling paradigms (a lexical baseline, a domain-pretrained encoder and a general-purpose large language model) classify financial headlines, and why do they differ? The second is conceptual: even a very accurate classifier measures only what a text says. Whether that matches what the market believes, whether the market has already reacted by the time the label exists, and how much each item should count are separate questions that benchmark accuracy cannot answer. The report addresses four questions:

- (1) How do TF-IDF + logistic regression, FinBERT and Qwen2.5-1.5B compare on financial headline classification, and why do their results differ?
- (2) In which application areas can the results be used, and with what limits?

- (3) Does headline sentiment faithfully reflect market attitude, does it arrive fast enough, and should every headline carry the same weight?
- (4) How could the original design be upgraded, given that FinBERT is an early BERT-based model and far stronger LLMs are now available?

Section 3 describes the data and the methodology. Sections 4 and 5 answer the first question, Section 6 the second, Section 7 the third and Section 8 the fourth.

2. Background and Related Work

Early financial text analysis relied on dictionaries. Loughran and McDonald (2011) showed that most words classed as negative by a general-purpose dictionary are not negative in financial filings, which motivated finance-specific word lists. Statistical classifiers such as TF-IDF with logistic regression learn word weights from labelled examples but remain bag-of-words models. Transformer encoders such as BERT (Devlin et al., 2019) represent words in context. FinBERT (Araci, 2019) adapts BERT to finance by further pretraining on financial text and fine-tuning it for sentiment, and it was benchmarked on the Financial PhraseBank (Malo et al., 2014). More recently, general-purpose LLMs can classify sentiment from a prompt alone, and Lopez-Lira and Tang (2023) report that LLM-generated headline scores predict subsequent stock returns, with stronger effects for small stocks and negative news.

A parallel literature asks what news sentiment means for prices. Tetlock (2007) found that media pessimism predicts downward pressure on prices followed by reversion, suggesting that sentiment mixes genuine information with temporary investor reaction. Later work shows that returns respond less to stale news although retail investors still trade on it (Tetlock, 2011), and that volatility and trading respond mainly to news identified as relevant to fundamentals (Groß-Klußmann & Hautsch, 2009; Boudoukh et al., 2019). These findings frame the critical discussion in Section 7.

3. Methodology

This section follows the order of the workflow in the Kaggle notebook (code: <https://www.kaggle.com/code/bennygao2/sentiment-analysis>): load and examine the data, split it, run the three models, then evaluate them and analyse the errors (Figure 1).

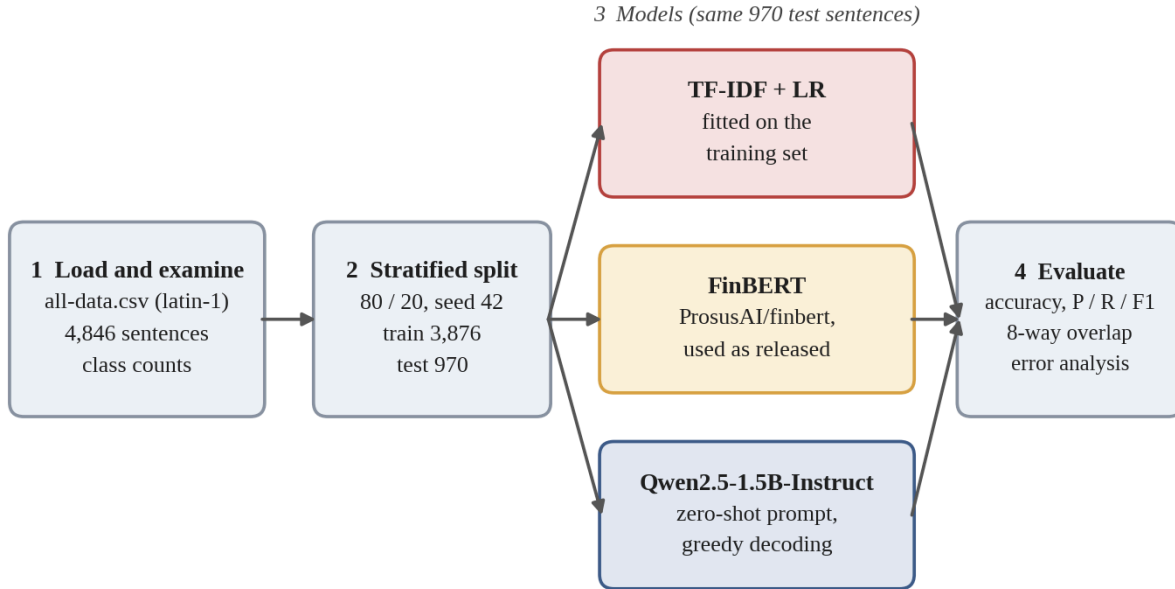


Figure 1. Workflow of the experiment. All three models are evaluated on the same 970 test sentences.

3.1 Dataset and loading

The data are the Kaggle dataset “Sentiment Analysis for Financial News” (<https://www.kaggle.com/datasets/ankurzing/sentiment-analysis-for-financial-news>), which distributes the 50%-agreement release of the Financial PhraseBank (Malo et al., 2014). The file `all-data.csv` has no header row and contains 4,846 rows with two fields, a sentiment label and a sentence. It was loaded with pandas, assigning the column names `sentiment` and `text` and reading the file with latin-1 encoding. Although the Kaggle column is called a headline, each item is a single sentence taken from financial news or company press releases about companies listed on the Helsinki stock exchange (OMX Helsinki), so the texts are short and self-contained but not strictly headlines; for simplicity they are still called headlines in this report. The file contains no dates, company identifiers or source names, which limits what can be analysed in Section 7. No manual cleaning was applied: all three models receive each sentence exactly as it appears in the file, including some garbled Nordic characters (for example, “Wärtsilä” appears as “W+Ærtsil+Æ”).

3.2 Data examination

Counting the labels with `df["sentiment"].value_counts()` gives 2,879 neutral, 1,363 positive and 604 negative sentences (Table 1 and Figure 2). Neutral is the majority class with 59.4% of the data, positive accounts for 28.1% and negative for 12.5%, so there are about 4.8 neutral sentences for every negative one.

Table 1. Class distribution of the dataset ($n = 4,846$).

Label	Count	Share of dataset
Neutral	2,879	59.4%
Positive	1,363	28.1%
Negative	604	12.5%
Total	4,846	100%

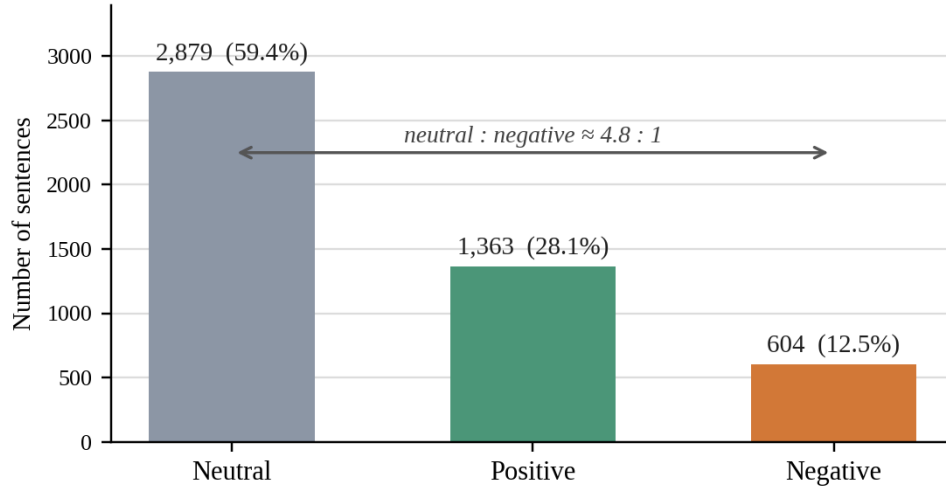


Figure 2. Class distribution of the 4,846 labelled sentences. Neutral outnumbers negative by about 4.8 to 1.

The labels follow a specific convention. Annotators judged each sentence from an investor’s point of view, that is, whether it could influence the company’s stock price, using only the information in the sentence; sentences without economic relevance count as neutral (Malo et al., 2014). Two properties of the labels matter for interpreting the results. First, they are not unanimous: the 50%-agreement release keeps sentences on which as few as half of the annotators agreed. Roughly 47% of the sentences are unanimous, 25% have 75–99% agreement, 16% have 66–74% and 13% have 50–65%. The Kaggle file does not include the agreement level, so this ambiguity cannot be examined row by row, but it implies that no model should be expected to reach 100%, and that disagreement with a label is not always a model error (Section 5.5). Second, all annotators came from one institution, so the labels reflect a small panel’s reading rather than a market consensus (Section 7.1).

3.3 Train–test split

The sentences were divided with `scikit-learn`’s `train_test_split` using `test_size=0.2`, `random_state=42` and `stratify` on the label. This gives 3,876 training sentences and 970 test sentences, and stratification keeps the class proportions in both parts: the test set contains 576 neutral (59.4%), 273 positive (28.1%) and 121 negative (12.5%) sentences. Only the TF-IDF + logistic regression model is trained on the training sentences. FinBERT and Qwen are not trained on this data, and all three models are scored on the same 970 test sentences so that their predictions can be compared one by one.

The test composition fixes a trivial baseline. A classifier that always answers “neutral” would be right on 576 of 970 sentences (59.4% accuracy), but its F1 would be about 0.75 on the neutral class and zero on the other two, so its Macro F1 would be only about 0.25 (Weighted F1 about 0.44). Accuracy is therefore a weak yardstick here, and the minority negative class, which matters most for risk applications, can be neglected at little cost in accuracy. Macro F1 and per-class recall are used alongside accuracy throughout, and Figure 3a adds this baseline to the results.

3.4 Model 1: TF-IDF + logistic regression

A `TfidfVectorizer` with default settings (lower-casing, single words, no stop-word removal and no limit on vocabulary size) was fitted on the training sentences with `fit_transform` and applied to the test sentences with `transform`, so that no information from the test set enters the vocabulary or the weights. The features were passed to `LogisticRegression(max_iter=1000)`, otherwise with default settings (L2 regularisation, $C = 1$). No class weights, resampling or hyperparameter tuning

were used, so the model is a deliberately simple baseline that inherits the class imbalance of the training data.

3.5 Model 2: FinBERT

FinBERT was loaded from the Hugging Face checkpoint `ProsusAI/finbert` with `AutoTokenizer` and `AutoModelForSequenceClassification` and used as released, without any further training. Each test sentence was tokenised separately (truncated at 512 tokens), passed through the model on a Tesla T4 GPU with gradients disabled, and the class with the highest score was converted to a label through the model’s own `id2label` mapping (positive, negative or neutral). The checkpoint is the released model of Araci (2019), which was fine-tuned on the Financial PhraseBank. Because this dataset is drawn from the same corpus, some test sentences may have been part of its training data (Section 8.1).

3.6 Model 3: Qwen2.5-1.5B-Instruct

Qwen2.5-1.5B-Instruct was loaded from `Qwen/Qwen2.5-1.5B-Instruct` in half precision (float16) on the same GPU and used zero-shot, that is, without examples or training. Each test sentence was inserted into the prompt below, which was sent as a single user message through the model’s chat template:

```
You are a financial sentiment classifier.

Classify the following financial news into exactly ONE of these three labels:
positive
neutral
negative

News:
{text}

Answer with only one label: positive, neutral, or negative.
```

Generation used greedy decoding (`do_sample=False`) and at most five new tokens, so the predictions are deterministic and reproducible. The reply was lower-cased and converted to a label by checking whether it contains “positive”, then “negative”, then “neutral”; any other reply was recorded as “unknown”, which counts as an error. The notebook does not record how many replies fell into this category.

Table 2 summarises the three configurations.

Table 2. *Configuration of the three models.*

Model	Implementation	Trained on this data?	Prediction rule
TF-IDF + LR	<code>TfidfVectorizer()</code> + <code>LogisticRegression(max_iter=1000)</code> ; scikit-learn defaults otherwise	Yes, on the 3,876 training sentences; no class weights or tuning	Class with the highest predicted probability
FinBERT	<code>ProsusAI/finbert</code> ; one sentence at a time on a Tesla T4 GPU; 512-token limit	No (used as released)	Highest score, mapped through the model’s label names

Model	Implementation	Trained on this data?	Prediction rule
Qwen2.5-1.5B	Qwen/Qwen2.5-1.5B-Instruct; float16; chat template; zero-shot prompt	No	Greedy decoding, at most 5 new tokens; keyword match on the reply, otherwise “unknown”

3.7 Evaluation and error analysis

Each model was scored with scikit-learn’s `accuracy_score` and `classification_report`, which give accuracy, per-class precision, recall and F1, and the macro and weighted averages. Macro F1 is the unweighted mean of the per-class F1 values, so the minority negative class counts as much as the neutral class. The predictions of the three models were then collected in one table together with the true label, and a correct/incorrect flag was computed for each model. Boolean masks on these flags partition the 970 test sentences into eight mutually exclusive groups (all correct, all wrong, each single model correct and each pair correct), and the sentences in each group were displayed and read to understand the error patterns (Sections 4.3 and 4.4).

4. Results

4.1 Overall performance

Table 3 and Figure 3 summarise overall performance. FinBERT classified 846 of 970 sentences correctly (87.22%), against 727 (74.95%) for both TF-IDF and Qwen: a margin of 119 sentences, or 12.27 percentage points. Relative to the always-neutral baseline (59.4%), TF-IDF and Qwen add about 15 percentage points of accuracy and FinBERT about 28. The gap is equally visible in Macro F1 (0.86 versus 0.72 and 0.67, against 0.25 for the baseline), which shows that FinBERT’s lead is not produced by the dominant neutral class. TF-IDF and Qwen tie on accuracy, but Qwen’s higher Macro and Weighted F1 indicate more balanced predictions across classes.

Table 3. *Model performance on the 970-sentence test set.*

Model	Accuracy (%)	Macro F1	Weighted F1
Always-neutral baseline	59.4	0.25	0.44
TF-IDF + LR	74.95	0.67	0.73
Qwen2.5-1.5B	74.95	0.72	0.75
FinBERT	87.22	0.86	0.87

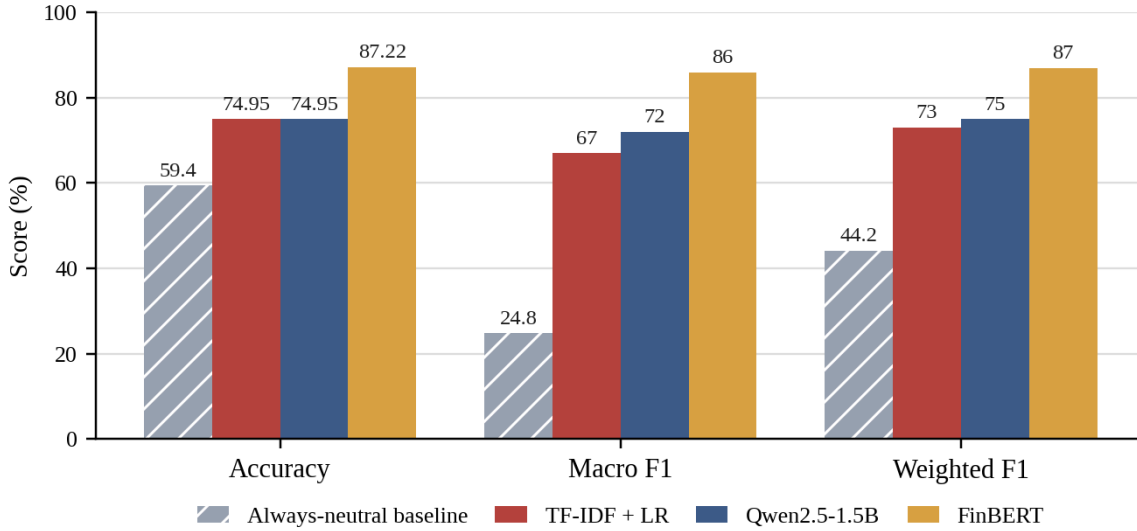


Figure 3. Accuracy, Macro F1 and Weighted F1 for each model and for an always-neutral baseline.

4.2 Per-class performance

Table 4 and Figure 4 break the results down by class. TF-IDF leans heavily towards the majority class: it recalls 92% of neutral sentences but only 46% of negative and 52% of positive ones. Qwen is the most cautious on negatives (precision 0.93 but recall 0.57) and the least precise on positives (0.65). FinBERT recalls 97% of negative and 90% of positive sentences, at the cost of the lowest precision on negatives (0.76) and a neutral recall (0.84) below TF-IDF’s (0.92). In other words, FinBERT trades some neutral recall for much better detection of sentiment-bearing sentences, the behaviour examined in Section 5. For any application that must not overlook bad news, the difference in negative recall matters more than the headline accuracy figure.

Table 4. Per-class precision (P), recall (R) and F1 on the test set.

Model	Negative (n = 121) P / R / F1	Neutral (n = 576) P / R / F1	Positive (n = 273) P / R / F1
TF-IDF + LR	0.80 / 0.46 / 0.59	0.76 / 0.92 / 0.83	0.71 / 0.52 / 0.60
Qwen2.5-1.5B	0.93 / 0.57 / 0.71	0.78 / 0.81 / 0.79	0.65 / 0.70 / 0.67
FinBERT	0.76 / 0.97 / 0.85	0.95 / 0.84 / 0.89	0.79 / 0.90 / 0.84

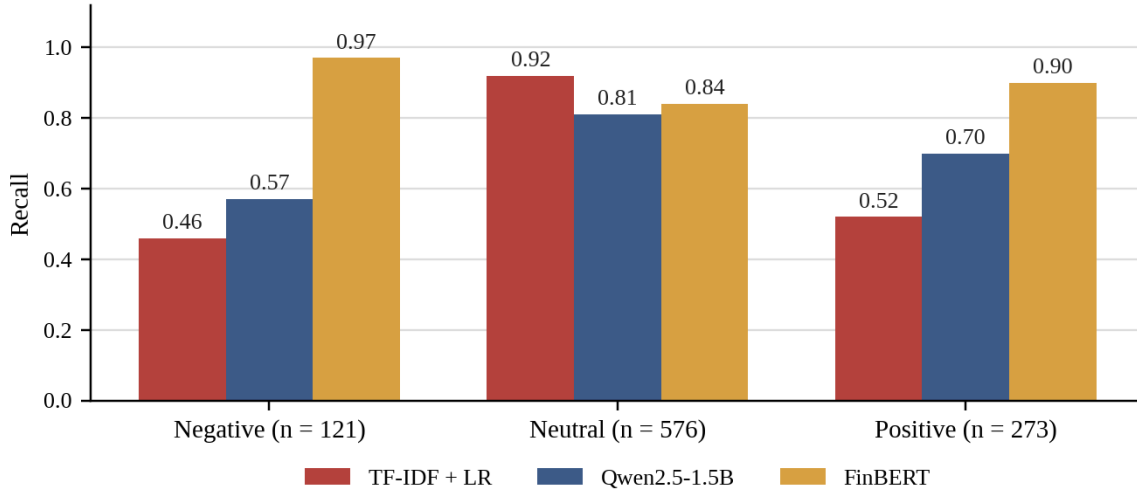


Figure 4. Recall by class. TF-IDF is strongest on the majority class, FinBERT on the two minority classes.

4.3 Case-level overlap

Accuracy tells us how often a model is right, not whether models are right on the same cases. Figure 5 partitions the 970 test sentences by which models were correct. All three were correct on 545 sentences (56.2%) and all three were wrong on 24 (2.5%). On the remaining 401 sentences (41.3%) only one or two models were correct, and FinBERT was among the correct ones in 301 of them (75%). FinBERT was the only correct model on 80 sentences, more than twice Qwen (34) and more than three times TF-IDF (23).

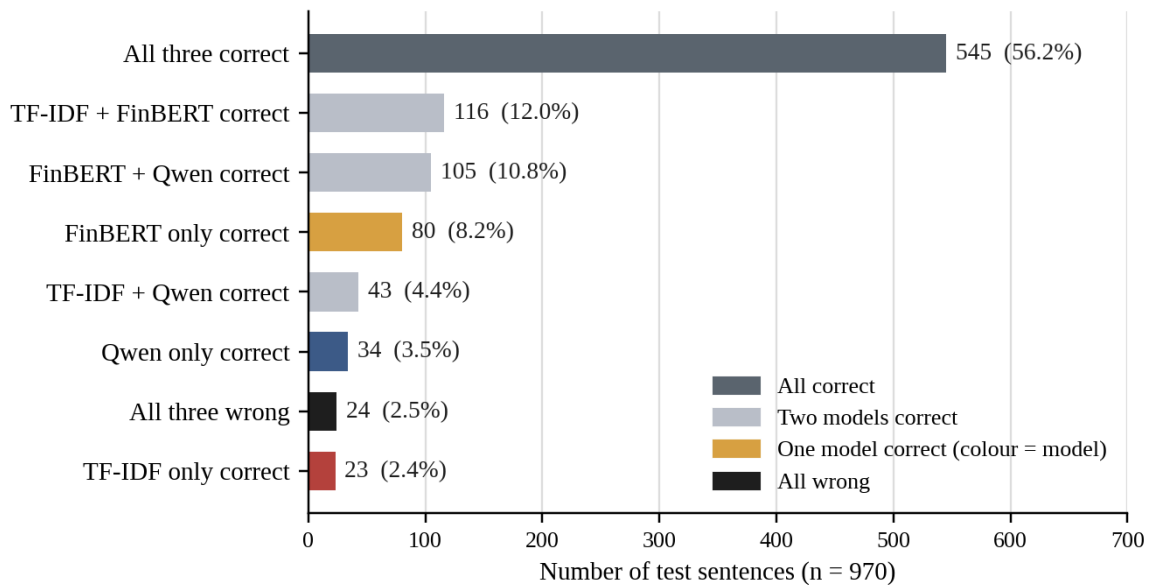


Figure 5. Case-level overlap: how many test sentences each combination of models classified correctly (the eight groups computed in the notebook).

Figure 6 shows the opposite measure: sentences on which a model was the only one to fail. FinBERT was the sole failure on 43 sentences, Qwen on 116 and TF-IDF on 105. Netting the two measures, FinBERT gains 37 sentences relative to the other models, whereas Qwen and TF-IDF each lose 82.

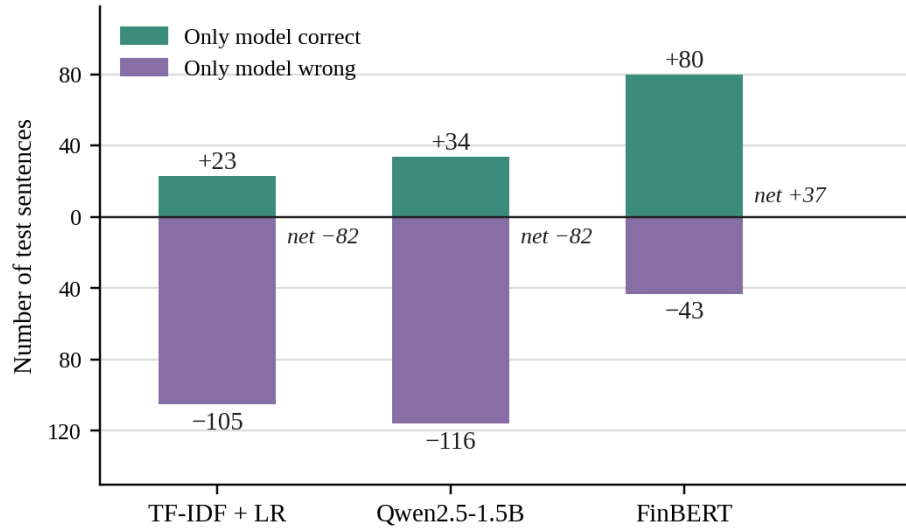


Figure 6. Sentences on which each model was the only one correct (above the axis) or the only one wrong (below the axis).

4.4 What the misclassified sentences look like

Reading the sentences in each group of the overlap analysis shows consistent patterns (examples in Table 5):

- **TF-IDF only correct (23).** All 23 sentences are neutral. FinBERT and Qwen made the same mistake on each of them, labelling 18 positive and 5 negative, typically because the sentence mentions a partnership, an expansion or a share-price move.
- **FinBERT only wrong (43).** 42 of the 43 sentences are neutral and one is positive. FinBERT labelled 27 of them positive and 16 negative, triggered by event words such as “shut”, “layoffs”, “toppled” or “fall” (negative) and “deal”, “agreement” or “inaugurated” (positive).
- **Qwen only correct (34).** 18 are neutral, 13 positive and 3 negative. They include numerical comparisons (“Return on investment was 16.6 % compared to 15...”, positive), a less obvious negative (“A tinyurl link takes users to a scamming site ...”) and price-move reports that TF-IDF and FinBERT read as negative.
- **FinBERT only correct (80).** The ten sentences displayed in the notebook turn on domain-specific polarity, such as “The diesel margin has remained high.” (positive) and “The measures result from weak demand in the sh...” (negative).
- **Qwen wrong, both others correct (116).** Of the ten examples displayed, six are neutral sentences that Qwen labelled positive and four are positive or negative sentences that it labelled neutral.
- **All three wrong (24).** 13 are positive, 10 neutral and only one negative: the models agree with each other but not with the label, at the boundary between positive and neutral.

Table 5. Illustrative sentences from the error analysis (text as displayed in the notebook, truncated where marked).

Group	Sentence	True	TF-IDF	FinBERT	Qwen
FinBERT only correct	“The diesel margin has remained high.”	pos	neu	pos	neu
FinBERT only correct	“The measures result from weak demand in the sh...”	neg	pos	neg	neu

Group	Sentence	True	TF-IDF	FinBERT	Qwen
TF-IDF only correct	“Garmin fell 4.5 percent to \$34.53 at 1:33 p.m...”	neu	neu	neg	neg
TF-IDF only correct	“Vacon aims to establish its presence in the so...”	neu	neu	pos	pos
Qwen only correct	“Return on investment was 16.6 % compared to 15...”	pos	neu	neg	pos
Qwen only correct	“A tinyurl link takes users to a scamming site ...”	neg	neu	neu	neg
Qwen only correct	“At 10.58 am, Outokumpu declined 2.74 pct to 2...”	neu	neg	neg	neu
FinBERT only wrong	“The company will also shut one paper machine i...”	neu	neu	neg	neu
FinBERT only wrong	“High winds also toppled three semi-trailers on...”	neu	neu	neg	neu
FinBERT only wrong	“The deal also includes a ten-year maintenance ...”	neu	neu	pos	neu
All three wrong	“Pre-tax loss totaled EUR 0.3 mn, compared to ...”	pos	neg	neg	neu
All three wrong	“We are also pleased to welcome the new employee...”	neu	pos	pos	pos

5. Discussion: Why the Models Differ

5.1 Different knowledge, different training

The models differ because each brings a different kind of knowledge to the task and is trained for it in a different way (Table 6). TF-IDF knows only which words co-occur with which labels in the training split, and that split is dominated by neutral sentences. Qwen knows a great deal about language, numbers and the world in general, but it has no financial prior and was never taught this dataset’s label convention, so its idea of what counts as “positive” is its own. FinBERT combines financial-domain pretraining with a classification head trained on labelled financial sentences, so both its vocabulary and its decision boundary are aligned with the task.

Table 6. *What each model has learned, and where it succeeds and fails.*

Model	What it has learned	Strengths	Weaknesses
TF-IDF + LR	Word-level statistics from the 3,876 training sentences, where the majority class dominates	Never attributes sentiment to events: all 23 sentences it alone got right are neutral; best neutral recall (0.92)	Polarity that depends on financial context; leans to neutral, so recall on negative (0.46) and positive (0.52) is low; lowest Macro F1 (0.67)
FinBERT	General language plus financial-domain pretraining, with a classification head trained for the task	Minority classes: recall 0.97 on negative and 0.90 on positive; domain language such as margins and weak demand (80 unique correct)	Over-interpretation: 42 of the 43 sentences on which only FinBERT failed are neutral; negative precision 0.76; neutral recall 0.84

Model	What it has learned	Strengths	Weaknesses
Qwen2.5-1.5B	Broad semantic and numerical knowledge; no financial prior and no training on this task	Numerical comparisons and unusual negatives (34 unique correct); more balanced than TF-IDF (Macro F1 0.72)	Lowest positive precision (0.65); wrong on 116 sentences that both others got right; cautious on negatives (recall 0.57)

5.2 Why FinBERT is strongest

FinBERT’s lead has two main sources. First, domain pretraining lets it read context-dependent financial language correctly. Among the sentences that only FinBERT classified correctly are “The diesel margin has remained high.” (positive) and “The measures result from weak demand in the sh...” (negative), where TF-IDF and Qwen gave neutral or even the opposite label. This also explains its high negative recall: negative financial news is often expressed through terms (falling margins, weak demand, tighter credit) whose polarity is specific to the domain. Second, FinBERT is a discriminative model trained for exactly this three-way decision, so its outputs match the dataset’s label space, whereas a prompted LLM has to infer that space from an instruction alone. A third factor must be kept in mind: the checkpoint was fine-tuned on the Financial PhraseBank, the corpus this dataset comes from, so part of FinBERT’s advantage may reflect exposure to the test sentences (Section 8.1).

5.3 FinBERT’s weakness: over-interpretation

FinBERT’s main weakness is the mirror image of its strength. Of the 43 sentences on which it was the only incorrect model, 42 are neutral in the dataset, and FinBERT labelled 27 of them positive and 16 negative. The triggers are mostly surface cues of an event rather than evidence about the company’s prospects: words such as “shut”, “layoffs”, “toppled” or “fall” push it towards negative, and “deal”, “agreement” or “inaugurated” towards positive (Table 5). This matters because the dataset’s convention treats an event with no clear economic implication as neutral (Section 3.2). The per-class figures show the same pattern: FinBERT’s negative precision (0.76) is the lowest of the three models and its neutral recall (0.84) is below TF-IDF’s (0.92). A model that is sensitive to financial language may treat every financially meaningful event as sentiment-bearing; financial relevance is not the same as financial sentiment. Share-price reports are a related case: sentences such as “Garmin fell 4.5 percent to \$34.53 ...” are labelled neutral in this dataset, yet FinBERT and Qwen both read them as negative.

5.4 Qwen and TF-IDF: equal accuracy, different behaviour

TF-IDF and Qwen reach the same accuracy for opposite reasons. TF-IDF, trained on imbalanced data without class weights, falls back on the majority class: it recalls 92% of neutral sentences but only 46% of negative and 52% of positive ones. Because it does not attribute sentiment to business events, it was the only correct model on 23 sentences, all of them neutral, on which FinBERT and Qwen made the same mistake, 18 times positive and 5 times negative.

Qwen is more balanced (Macro F1 0.72 versus 0.67) and was uniquely correct on 34 sentences: 18 neutral, 13 positive and 3 negative, including numerical comparisons such as return on investment and a link to a scamming site. However, in 116 cases where both other models were correct, Qwen failed. Of the ten examples displayed in the notebook, six are neutral sentences that it labelled positive, which fits its low precision on positive sentences (0.65) and suggests a bias towards associating business activity with favourable outcomes. On negatives it is cautious: precision 0.93, but recall only 0.57.

5.5 Complementarity and label noise

A large share of the data (545 sentences, 56.2%) is classified correctly by every model, while 401 (41.3%) were solved by only some of them. This complementarity is useful for system design (Section 8). All three models failed on 24 sentences, and their composition is telling: 13 are positive, 10 neutral and only one negative. The negative class is almost never missed by every model; the difficulty lies at the boundary between positive and neutral, where the three models usually agree with each other but not with the label. This fits the label ambiguity described in Section 3.2, since only about 47% of the corpus is unanimously labelled. Some of the cases are also genuine comparison problems: a pre-tax loss reported “compared to” an earlier figure is labelled positive, presumably because the loss narrowed, while TF-IDF and FinBERT said negative and Qwen neutral. Errors on such sentences should therefore not automatically be read as model failures; a manual audit of this group would show how much of the apparent error ceiling is label noise.

6. Application Areas

The three models suggest different roles, and the right choice depends on the cost of each type of error (Table 7).

Table 7. *Application areas, the metric that matters most, and the main risk.*

Application	Typical use	Metric that matters most	Main risk
Investment research	News monitoring, sentiment dashboards, screening for negative developments	Recall on the negative class	Over-interpreted neutral events create alert fatigue
Risk management and compliance	Early warning on credit, supply-chain and reputational news	Negative recall; low false-negative rate	Missed negatives leave exposures undetected (Qwen recall 0.57)
Corporate and customer analytics	Earnings-call, analyst-note and customer-feedback analysis	Balanced Macro F1; quality of explanations	Domain shift from short sentences to long documents
Quantitative signals	Sentiment features for trading and forecasting models	Precision on positive and negative calls; market-linked performance	False signals from neutral events; sentiment is not a price move (Section 7)

Two points follow. First, the metric that matters differs by application. Early-warning uses depend on negative-class recall, which favours FinBERT. Signals that feed a trading model depend on precision, where FinBERT’s over-interpretation of neutral events would produce false positives unless a separate step distinguishes factual events from sentiment. Second, all of these applications assume that headline sentiment is a meaningful signal about the market. Section 7 tests that assumption.

7. Does Sentiment Analysis Reflect the Market?

A classifier that agrees with human annotators on 87% of sentences has shown that it can read headlines the way those annotators did. It has not shown that it measures the market. Figure 7 breaks the path from a real-world event to a market reaction into steps and marks three places where information is lost or distorted. The rest of this section examines each in turn.

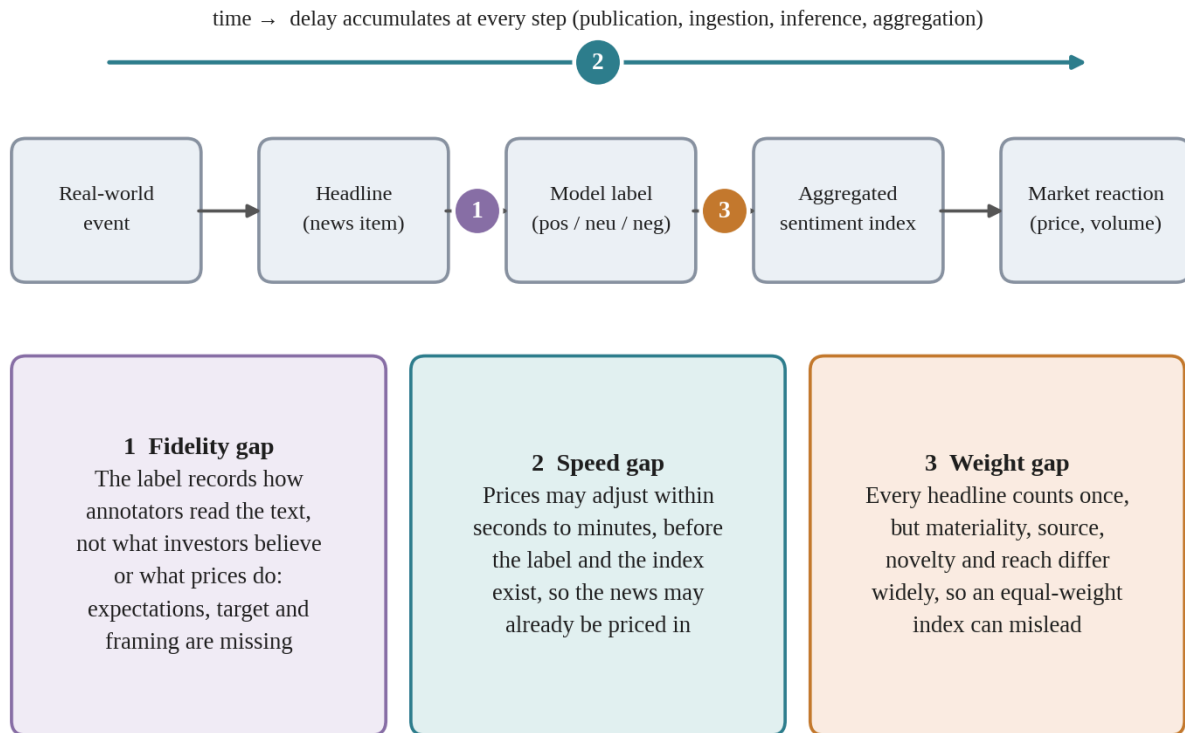


Figure 7. From event to market reaction: three gaps between a sentiment label and market behaviour.

7.1 Fidelity: does the label reflect market attitude?

“Market attitude” can mean three different things: the tone of a text, the beliefs of investors, and the behaviour of prices and volumes. The benchmark measures agreement with the first, as judged by annotators. According to the dataset documentation, the Financial PhraseBank labels come from 16 annotators (three researchers and 13 master’s students at one business school) who judged each sentence in isolation, and the documentation itself cautions that inter-annotator agreement should be read with this shared institution in mind. Agreement with a small panel is not agreement with the market.

Four further problems follow from the way a headline is turned into a label. First, sentiment depends on expectations: a 10% fall in profit is good news if analysts expected a 20% fall, but a classifier that sees only the sentence measures the level of the news, not the surprise. Second, polarity depends on the target: “oil prices climb” is good for a producer and bad for an airline, yet a three-class label assigns a single polarity to the sentence. Third, headlines are selected and compressed by editors, and many describe price moves that have already happened (“shares jump after results”), so the sentiment may follow the price instead of anticipating it. Fourth, there is no single market attitude: retail and institutional investors, long and short holders, interpret the same news differently.

The evidence on predictive power is real but conditional. Tetlock (2007) found that media pessimism predicts downward pressure on prices followed by reversion, so sentiment mixes information with temporary investor reaction. Lopez-Lira and Tang (2023) report that LLM-based headline scores predict subsequent returns, particularly for small stocks and negative news, and that strategy returns decline as LLM adoption rises. Sentiment therefore carries information, but its link to prices varies across stocks and over time and may erode as more participants use it. The 87% figure should be read as “labels sentences like an investor-minded annotator”, not as “measures the market”.

7.2 Speed: has the market already reacted?

Under the efficient-market view (Fama, 1970), public information is incorporated into prices quickly, and high-frequency evidence is consistent with this. Groß-Klußmann and Hautsch (2009) measured reactions to machine-classified news at 20-second intervals and found responses in volatility and trading volume, but only for news classified as highly relevant. A headline that a pipeline classifies minutes or hours after publication, and a daily aggregate even more so, has often been priced in by the time the label exists. Every step in Figure 7 adds delay: ingestion, inference and aggregation. Small encoders typically classify in milliseconds on a GPU, whereas large LLMs reached through an API are usually slower, which matters for intraday use but little for daily monitoring.

Speed therefore matters differently by horizon and by stock. Figure 8 illustrates the point with stylised curves. For a liquid, heavily covered stock most of the price adjustment is complete within minutes, long before a daily sentiment output exists. For an illiquid, low-attention stock, adjustment is slower and a daily signal may still arrive during the process. This is consistent with the drift after news that Lopez-Lira and Tang (2023) report for small stocks and negative news, and with the reversal documented by Tetlock (2007). The value of a sentiment signal thus lies less in beating the first reaction than in capturing underreaction, overreaction and risk context at horizons of days.

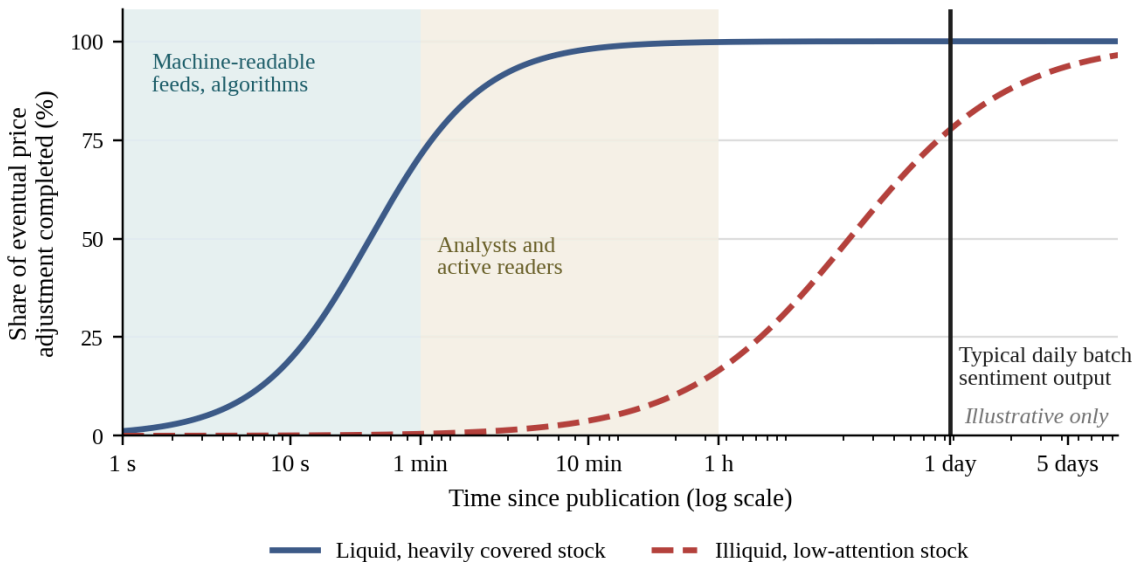


Figure 8. Stylised timing of price adjustment after a news item. *Illustrative only: the curves and time windows are assumptions for discussion, not estimates from this project's data.*

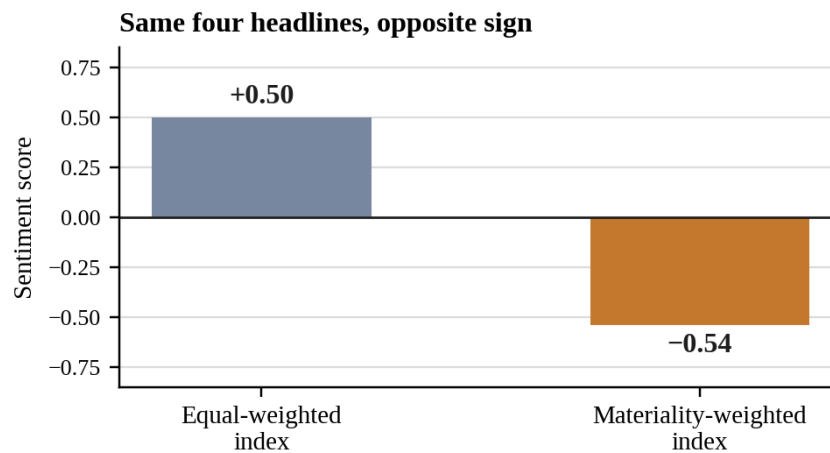
The dataset used here contains no timestamps, tickers or prices, so none of these timing questions can be tested with it. Section 8 proposes a market-linked extension in which each headline is matched to its publication time and the subsequent return path.

7.3 Weight: does every headline count equally?

The benchmark gives every sentence the same weight: accuracy counts each correct label once, whatever the sentence is about. In practice, errors do not cost the same. Labelling a profit warning as neutral is far more costly than mislabelling a routine product update, and the same asymmetry applies when labels are aggregated into an index. Table 8 gives a hypothetical illustration: three minor positive items and one major negative item produce a positive equal-weighted index but a negative materiality-weighted index (Figure 9).

Table 8. Hypothetical example: the same four headlines under equal and materiality weighting.

Headline (hypothetical)	Label (score)	Materiality weight	Weighted score
Firm launches a customer loyalty app	Positive (+1)	0.1	+0.1
Firm opens a new regional office	Positive (+1)	0.1	+0.1
Firm announces a minor product update	Positive (+1)	0.1	+0.1
Firm cuts full-year profit guidance	Negative (−1)	1.0	−1.0
Equal-weighted index = $(+1 + 1 + 1 - 1) / 4$			+0.50
Materiality-weighted index = $(0.1 + 0.1 + 0.1 - 1.0) / 1.3$			−0.54

**Figure 9.** Equal versus materiality weighting for the headlines in Table 8 (hypothetical).

Real markets weight news along several dimensions, and the literature documents each:

- **Materiality.** Boudoukh et al. (2019) find that news identified as relevant to fundamentals is strongly linked to stock price changes, while many business news stories are not; Groß-Klußmann and Hautsch (2009) likewise find volatility and volume reactions only for highly relevant items.
- **Novelty.** Tetlock (2011) measures the staleness of a story as its textual similarity to the previous ten stories about the firm. Returns respond less to stale news, but retail investors still trade on it and the resulting price moves partly reverse.
- **Attention and reach.** Barber and Odean (2008) show that individual investors are drawn to attention-grabbing stocks, including those in the news, and Lopez-Lira and Tang (2023) find stronger predictability for small stocks. The same headline therefore has a different weight for different firms.
- **Timing.** Boudoukh et al. (2019) report that identified news accounts for about half of overnight idiosyncratic volatility (49.6%) but a much smaller share during trading hours (12.4%), so the weight of news depends on when it arrives.
- **Source and context.** Source credibility, duplication across outlets and the market regime are further sources of variation that headline-level labels ignore.

Weights should therefore be estimated rather than assumed, for example by regressing subsequent abnormal returns on sentiment interacted with materiality, novelty, source, firm size and timing. The

classification metric should also be cost-sensitive, penalising a missed negative on a material event more heavily than a confused neutral sentence.

7.4 Synthesis

None of this makes sentiment analysis useless: the literature cited above suggests that news text contains price-relevant information. The conclusion is narrower. Sentiment scores are best treated as one noisy, delayed and unequally informative input whose value has to be tested against market data for the intended horizon and stock universe, rather than as a direct measurement of market attitude. Table 9 summarises the three challenges and how each could be tested.

Table 9. *Three challenges to sentiment analysis as a market measure.*

Challenge	Core problem	Evidence	How to test
Fidelity	Labels record how annotators read a sentence, not investors' beliefs or price moves; expectations, target and framing are missing	Labels from 16 annotators at one institution (Malo et al., 2014); pessimism followed by reversal (Tetlock, 2007)	Link headlines to returns; target-dependent labelling; expectation-adjusted (surprise) features
Speed	Prices may adjust within seconds to minutes; batch pipelines arrive late	Reactions measured at 20-second intervals (Groß-Klußmann & Hautsch, 2009); drift for small stocks and negative news (Lopez-Lira & Tang, 2023)	Event study at several horizons; lead-lag analysis; measure pipeline latency
Weight	One headline is not one unit of information	Relevant news drives volatility (Boudoukh et al., 2019); stale news (Tetlock, 2011); attention (Barber & Odean, 2008)	Materiality, novelty and source models; learned weights; cost-sensitive metrics

8. Limitations and Proposed Upgrades

8.1 Limitations of the current design

- **Dated and small models.** FinBERT is an early BERT-based model and Qwen2.5-1.5B is small, so the comparison is between a domain-adapted small model and an untuned general small model. It says little about current frontier LLMs.
- **Untuned lexical baseline.** TF-IDF + logistic regression ran with default settings and no class weights, so its weak minority-class recall partly reflects the class imbalance. A tuned baseline (class weights, word pairs, regularisation chosen by cross-validation) would be a fairer yardstick.
- **One prompt, one run for Qwen.** Qwen was prompted once, without examples, with greedy decoding, and the number of unparseable replies was not recorded. Part of its disadvantage may reflect the prompt and the absence of task fine-tuning rather than the absence of domain knowledge.
- **Single split.** One stratified split (seed 42), without confidence intervals or a paired significance test, cannot show how stable the differences are, although the 119-sentence margin between FinBERT and the others is large.
- **Possible overlap with FinBERT's training data.** The checkpoint ProsusAI/finbert was fine-tuned on the Financial PhraseBank (Araci, 2019), which is the source of this dataset, and

the test split was drawn at random from the full dataset. Part of the test set may therefore have been seen during FinBERT’s training, which would overstate its advantage.

- **Narrow data and label convention.** The sentences come from news about companies listed in Helsinki, labelled by a small annotator panel under an investor-viewpoint convention, and the 24 all-wrong sentences suggest some ambiguity. Performance on other markets, periods or text types is untested.
- **No market data.** Without timestamps, tickers and prices, none of the questions in Section 7 can be tested.

8.2 Proposed upgraded design

The original comparison is useful, but it pits a 2019-era domain model against a small general model. Current frontier LLMs are far more capable. Lopez-Lira and Tang (2023) report that basic models such as BERT could not forecast returns accurately while forecasting ability grew with model size; their task, return prediction, differs from three-class labelling, but the result supports testing frontier models here. Table 10 sets out a six-part upgrade.

Table 10. *Proposed experiments for the upgraded project.*

Experiment	Design	Question it answers
A. Frontier LLM baselines	Claude Opus 5.5, Fable 5 and GPT-6.1; few-shot prompts, temperature 0, structured output (label, short rationale, confidence); repeated runs for stability; unparseable replies, cost and latency recorded	Does stronger general reasoning remove the positivity bias on neutral events, and does it close or reverse the gap to FinBERT?
B. Fair baselines	Tuned lexical baseline (class-weighted logistic regression, word pairs, regularisation by cross-validation); LoRA fine-tuning (Hu et al., 2022) of Qwen2.5 at several sizes (1.5B, 7B, 14B); DeBERTa-v3 and ModernBERT fine-tuned on the same training split	How much of FinBERT’s lead comes from domain pretraining, how much from task fine-tuning, scale and class handling?
C. Convention-aware prompting	State the dataset’s convention (investor viewpoint, only the information in the sentence, neutral if no economic implication) and include neutral-event examples	Does explicit label guidance reduce over-interpretation of neutral events?
D. Hybrid routing	Fine-tuned encoder first; escalate low-confidence cases to an LLM (Figure 10); threshold chosen on a validation set	Can accuracy approach LLM level at a fraction of the cost and latency?
E. Market-linked evaluation	Timestamped headlines with tickers and prices; event studies of abnormal returns over minutes, one day and five days; lead–lag analysis; equal-weighted versus learned-weight indices (materiality, novelty, source, firm size); target-dependent sentiment via structured LLM output	Does sentiment reflect market behaviour, how quickly, and does weighting improve it? (Section 7)
F. Rigorous statistics	McNemar’s test on paired errors, bootstrap confidence intervals, calibration, repeated stratified splits with different seeds, an external dataset that FinBERT cannot have seen (for example FiQA; Maia et al., 2018), and a manual audit of the 24 all-wrong sentences	Are the differences significant, robust, calibrated and free of leakage?

Figure 10 shows the hybrid design in Experiment D. Because 401 sentences were correct for only one or two models, the models are complementary. A cheap, fast encoder handles confident cases; low-confidence or conflicting cases are escalated to a frontier LLM, whose instruction-following and reasoning matter most on ambiguous, context-dependent sentences. Disagreements and all-wrong cases feed a manual annotation review, which in turn improves the prompt and the training data.

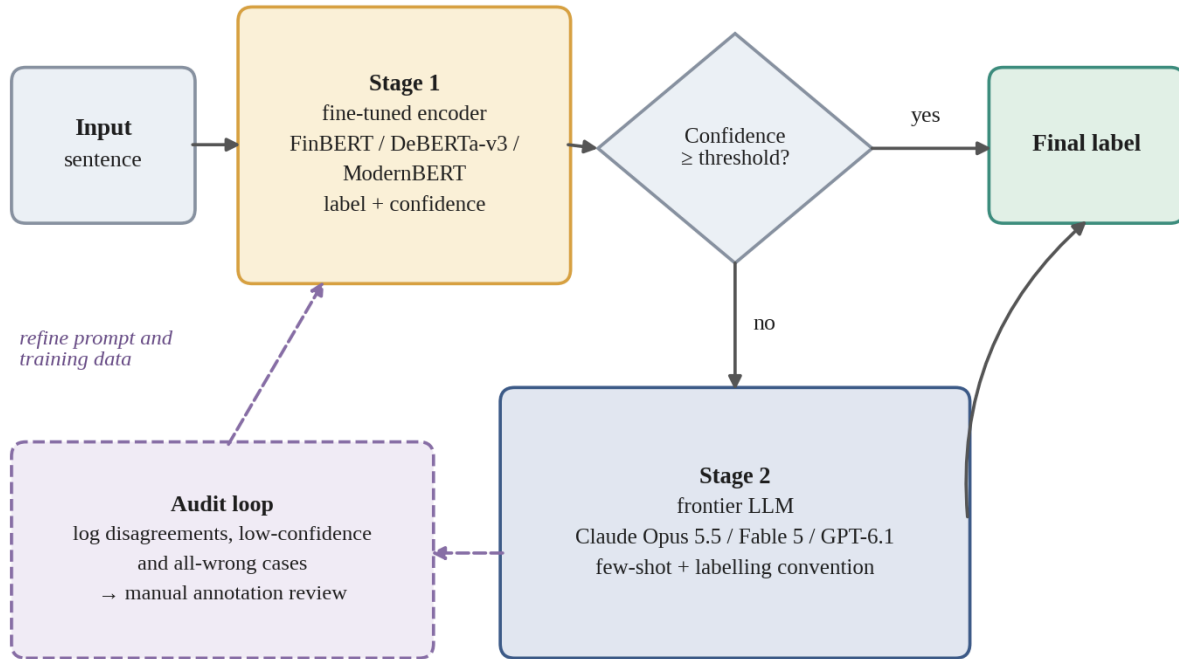


Figure 10. Proposed hybrid pipeline with confidence-based escalation and an audit loop.

These proposals are hypotheses rather than results. I expect frontier LLMs to handle context and numerical reasoning better than Qwen2.5-1.5B, but they may still label neutral business events as positive unless the labelling convention is stated; fine-tuned modern encoders should close much of the gap to FinBERT cheaply; and hybrid routing should approach the best single model’s accuracy at lower cost. For Experiment E, the prior literature suggests that a weighted index should outperform an equal-weighted one and that predictability should be stronger for small stocks and negative news, but whether this holds for a given sample and period has to be measured.

9. Conclusion

The data examination showed a dataset dominated by neutral sentences (59%), with a small negative class (12%), so accuracy alone is a weak guide and a trivial baseline already reaches 59%. On the 970-sentence test set, FinBERT clearly outperformed TF-IDF + logistic regression and Qwen2.5-1.5B (accuracy 87.22% against 74.95%; Macro F1 0.86 against 0.72 and 0.67), mainly through financial-domain pretraining and a task-specific classifier, although exposure to the Financial PhraseBank during its training may inflate the margin. Each model nevertheless showed distinct strengths and weaknesses: TF-IDF fell back on the majority class but never read sentiment into neutral events, Qwen was balanced but biased towards positive readings of business news, and FinBERT detected sentiment best but over-interpreted events (42 of its 43 unique errors were neutral sentences). The results are directly useful for early-warning and monitoring applications that depend on negative-class recall.

The more important lesson concerns what the benchmark measures. Accuracy shows agreement with annotators, not with the market. Whether sentiment reflects market attitude, arrives before prices have moved and deserves equal weight are separate empirical questions, and the literature suggests that each has a qualified answer. A stronger project would therefore add frontier LLMs and fair fine-tuned baselines, route cases between cheap and powerful models and, above all, link headlines to timestamps and prices so that sentiment can be judged by what the market actually does.

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